Comparison between conventional and solid state transformers in smart distribution grids

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***Abstract*—Due to the increasing insertion of distributed generation in the electric power system, new technologies are necessary to ensure power quality the energy quality. The concept of Smart Grids stands out because it provides an improvement by integrating distributed generation and control systems capable of communicating with each other. As a tool to make Smart Grids viable, the Solid State Transformer (SST) has been studied, capable to perform the same functions as the conventional distribution transformer (LFT – Low Frequency Transformer), as well as several auxiliary services that allow the required levels of quality to be achieved. Therefore, this work presents a review and a comparison between the conventional transformers and the SST, in order to highlight the importance of its implementation. In addition, the impact of SST in the power network instead of LFT is presented. Finally, the reactive power control results are presented for voltage and power factor regulation.**

***Index Terms*—Smart Grids, Solid State Transformer, SST, Energy Quality, Conventional Transformer**

I. Introduction

The complexity of the electrical power grid has been growing continuously, due to the increase in the use of renewable energies and distributed generation (DG), necessitating a bidirectional power flow in the distribution system, distributed control and management capacity, monitoring and self-healing [1]. From this scenario, the concept Smart Grids was proposed, which aims to mitigate or avoid the consequences of events related to energy quality (such as voltage sags), improve reliability (reducing the number and duration of interruptions) and increasing efficiency by using new technologies and equipment [2].

For this to be implemented, there is a need for the use of intelligent equipment that can communicate and be remotely controlled. One of the solutions is the Solid State Transformer (SST), which is able to perform the same functions of the conventional distribution transformer. In addition, the SST performs auxiliary services capable to improve and allow the connection between direct current (dc) and alternating current (ac) equipment.

There is a lot of studies about SST applications in power grids. In [3], several investigations are carried out in order to prove some of the SST functionalities such as voltage control, reactive power compensation and harmonic mitigation in the medium voltage, as well as power factor correction in low voltage. In addition, the mentioned functionalities are also approached by [1], however they present several scenarios and their respective results regarding harmonic mitigation, short-term disturbance, power factor correction, reactive compensation and bidirectional flow.

Another point that the SST stands out is the integration of DG, energy storage systems and electric vehicles. A way of electric vehicle charging integrated to the SST and photovoltaic charging stations is presented by [4], where SST allows better control of the charging process. In [5], the SST is presented as a way to optimize the integration of energy storage systems with the distribution network, evaluating its performance in relation to the conventional system. In [3] and [6], the SST is analyzed integrated to photovoltaic systems.

This paper implement the substitution of a LFT by an SST in order to check its advantages and disadvantages, as well as to make use of its capacity of reactive power control for power factor correction and voltage regulation. This control is performed through the volt/var strategy, where a fixed increment of reactive power is defined in order to increase or decrease the reactive power absorbed/injected by the SST.

II. SST Fundamentals

The Solid State Transformer is an electronic equipment composed of converters divided into 3 parts: a Medium Voltage (MV) stage, an isolation stage through a high frequency transformer and a Low Voltage (LV) stage.

The converter connected to the MV network converts the AC voltages of the three-phase supply frequency (50/60Hz) to a DC voltage, then converted back to ac, however at a high frequency. Due to the high frequency in ac, the core magnetic properties of the high frequency transformer are best used and the transformer can be considerably smaller while maintaining the same power capacity. After passing through the high-frequency transformer, the AC voltage is converted to dc, and then converted to AC with a frequency of 50/60 Hz [2].

The SST is a device that has the proposal to meet the demand of the smart grids. The Fig. 1 illustrates the application of the SST in the distribution network where it is able

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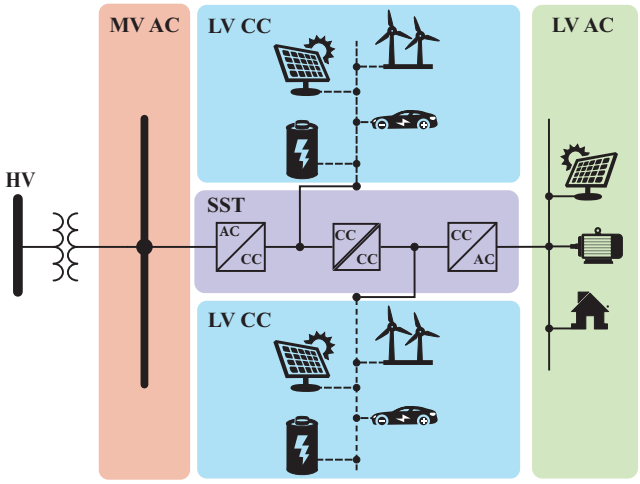


Fig. 1. SST applied to the distribution network

to replace the conventional distribution transformer in the connection of LV AC loads, and also allows the integration of directly connected DGs to its DC bus.

In addition to having all the functionalities of the conventional transformer, the main benefits of SST are: reduced size; allows bidirectional flow; reactive power compensation; surge protection, voltage sags, harmonics; voltage regulation; DC connectivity; reduction of short-circuit currents [2], [6]–[8].

Despite the great benefits of using the SSTs, these have disadvantages still not overcome in the current scenario. The SSTs have the following cons [9]: they are less efficient than the LFTs; have higher costs (5x more expensive than LFTs); are not compatible with current protection schemes; and are less robust.

III. Analysis Methodology

A. SST Modeling

The proposed modeling considers the SST representation as an equivalent circuit of an AC-AC converter, where the apparent power injected into the electric network, after the inverter filter terminals, is assumed to be $|S|e^{-j\theta r}$, where θr is the angle of the network power factor. Thereby, it is possible to define the limits of active and reactive power injection in the electric network, without considering limitations imposed by standards, and to establish a PQ plan, in order to represent all the points of operation of the SST. In addition, it is possible to define operating quadrants, which define the behavior of the SST in the network, i.e, depending on the operating point, it can act by consuming or injecting active power, as well as being able to operate decreasing or increasing voltage and correcting the power factor at the point of installation.

For photovoltaic (PV) systems, NBR 16149-2014 imposes limits of apparent power injected into the power grid in order to guarantee a power factor of 0.9. Due to the SST being relatively new equipment there are still no regulatory standards for the connection of this equipment to the network. In this way, the limits of NBR 16149 will be considered, where the SST can only inject reactive power when the active power injected is at least 20% of the nominal active power (Fig. 2).

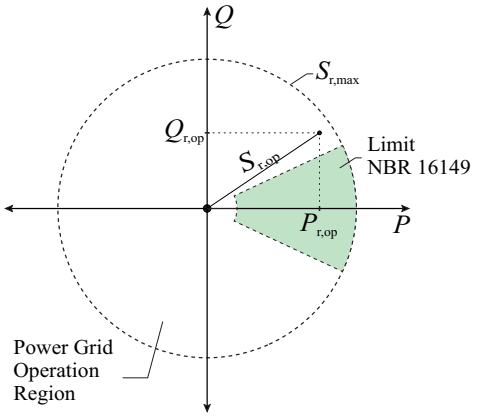


Fig. 2. PQ Plan

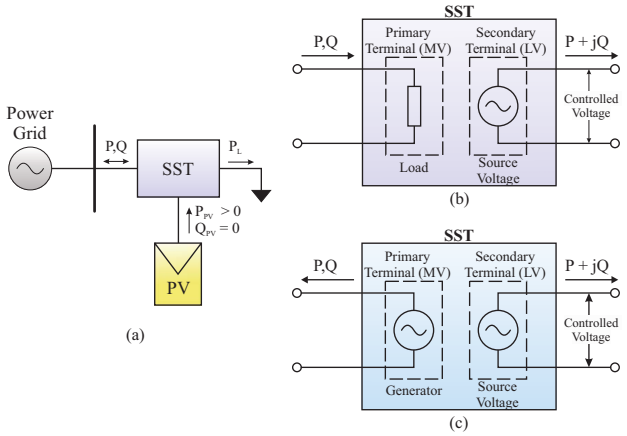


Fig. 3. SST modeling in OpenDSS®. (a) SST connection way with PV. (b) SST model operating as load. (c) SST model operating as generator.

B. OpenDSS® Modeling

The SST modeling is based on the model proposed by [2], where the SST is modeled as a two-terminal element. Thus, the LV side provides active and reactive power for the LV loads, while maintaining the voltage at a constant value at its secondary terminal (1 p.u). On the MV side, the SST absorbs only active power. As it can move in both directions of the SST, it can be configured in two modes of operation, (i) Load Mode and (ii) Generation Mode, shown in Fig. 3 (b) and Fig. 3(c), respectively.

The Fig. 3 shows how the SST is modeled in OpenDSS®. Since the SST allows the connection of DG, the model must contemplate this functionality, then Fig. 3(a) presents the power flow through the SST considering the connection of a PV which provides only active power according to the irradiance and temperature curves. The power flow between the network and the SST is determined based on the power generated by the PV (P_{PV}) and by the load demand (P_L).

When the SST is operating on Load Mode ($P_{PV} < P_L$), it will be modeled as a voltage source on LV side and a load of the PQ model in MV side. The voltage source provides all

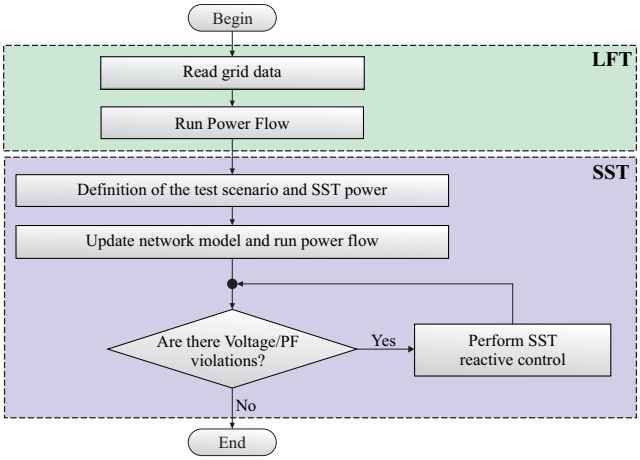


Fig. 4. Methodology Flowchart

the active and reactive power required by the loads in LV and keeps the voltage constant. The load replicates the behavior of the SST seen on the MV side, i.e, the active power demanded by the load in the MV side will be equal to the active power supplied by the source in LV side plus the losses of the SST.

When operating in Generation Mode ($P_{PV} > P_L$), the SST will be modeled as a voltage source on LV side and a generator on MV side, with fixed active power. In this way, the power flow reverses, injecting power into the MV network, with the active power provided by the LV side minus the SST losses.

C. Methodology Architecture

The Fig. 4 shows the flowchart of the architecture used. In the first stage, the distribution grid data will be obtained without SSTs, by reading the parameters of the network and the power flow. In the second stage, the test scenario will be defined, allocating SSTs in the grid, as well as the definition of the PQ powers of these. Then, the power flow is executed and it verifies if there are voltage and power factor violations, if it does, make the adjustments through the volt/var control.

D. SST Reactive Power Control

The volt/var control strategy was used to control the reactive power on the MV side of the SSTs. Basically, the control consists in defining a step of increasing or decreasing reactive power. The control of reactive power is performed by (1).

$$Q_{\text{SST}} = Q_{\text{SST}} \pm \Delta Q \tag{1}$$

where, Q_{SST} is the SST reactive power and ΔQ is the fixed step.

If there is a violation of the substation power factor, it is verified whether the substation reactive power is positive or negative, if positive the decrease of the SST reactive power is performed, otherwise the increment is performed. Regarding the voltage, if it violates the lower limit, the SST reactive power must be reduced in order to inject reactive power and, thus, raising the voltage. If the voltage violates the upper limit, the SST reactive power must be increased, causing it to absorb power, reducing the voltage.

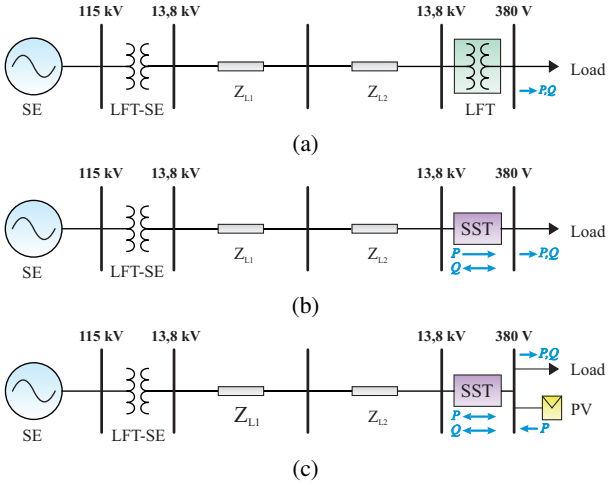


Fig. 5. Example system. (a) System using a LFT at the load. (b) System using an SST at the load. (c) System using SST and PV.

IV. Results

In order to demonstrate the functionality of the SST, a simple radial distribution system (Fig. 5) was designed using the OpenDSS® software integrated with the Python programming language. This system consists of a substation operating at 115 kV, a transformer (115/13.8 kV, $\Delta - \Delta$, 1000 kVA), two lines with 40 km length each, using "2_CA" conductor, with impedance of $0.963 + j0.987\ \Omega/km$ and a 200 kVA PQ load, connected at 380 V.

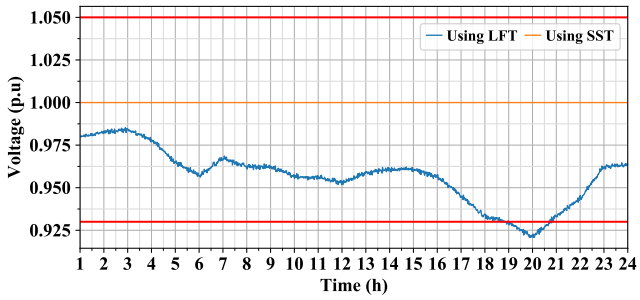
Initially, the connection of the load with the system is carried out by a LFT (13.8/0.22 kV, 200 kVA), according to Fig. 5(a), and, in order to compare the results, the conventional transformer is replaced by an SST, according to Fig. 5(b). Finally, the simulation is performed considering the insertion of a PV system connected on LV bus, in order to analyze the behavior of the SST when attached to this kind of system.

The elements have been defined in order to obtain a sufficient voltage drop in the load to violate the voltage limit of 0.93 p.u, as defined by the PRODIST - Module 8 [10].

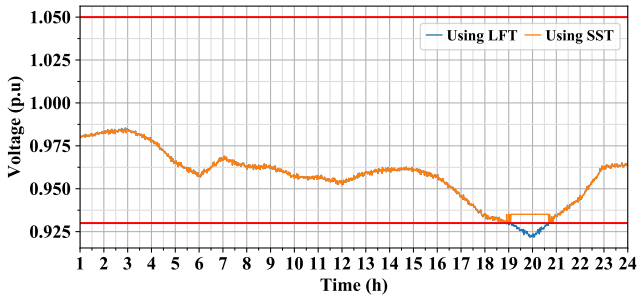
The load considered is a 200 kVA three-phase load, modeled as constant power and connected in Y with a line voltage of 380 V, which has a residential characteristic.

Regarding SST, since it must replace LFT, it is modeled by two elements (*Load* and *VSource*). The *VSource* is connected to the desired load bar, with the same line voltage and phase number parameters. The SST *Load* is connected to the MV bus, with the same characteristics of the load in LV, except for the line voltage which becomes the network voltage in MV (in this case 13.8 kV), reactive power (the SST will not consume reactive power, since the reactive power in LV is supplied by the internal capacitors) and the load curve, which will be adjusted according to the need of the load in LV.

All the simulations were performed in a 24-hour period, with 1 minute step, totaling 60 points per hour and 1440 points in the day, and the analyzes were based on three tests: (a) load

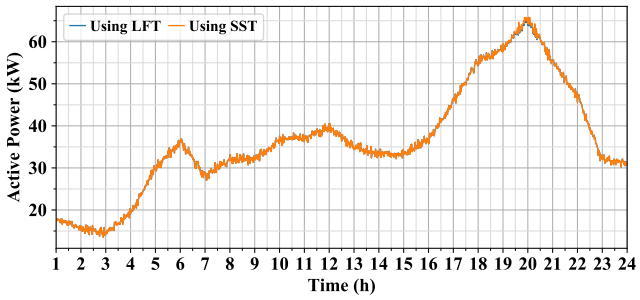


(a)

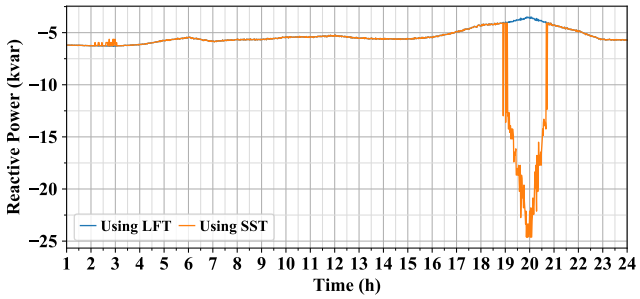


(b)

Fig. 6. Load Voltage. (a) Low Voltage. (b) Medium Voltage.



(a)



(b)

Fig. 7. Substation. (a) Active Power. (b) Reactive Power.

with unit power factor; (b) load with a power factor of 0.9 and (c) load with a power factor of 0.9 together with a PV system.

A. *Test 1*

The first test consists of using a purely resistive load, which only requires active power of the system in order to verify with the SST behaves as well as the operation of voltage and power factor control.

The Fig. 6(a) shows the voltage of the load in LV when a LFT is used and when an SST is used. The red lines indicate the voltage limits, according to [10], with the lower limit of 0.93 p.u and the upper limit of 1.05 p.u. The blue curve shows the load voltage in LV, which varies according to the load curve, showing a voltage level violation (V<0.93 p.u) in the period from 7:00 p.m. to 8:30 p.m. This behavior is solved by using an SST, since it guarantees the control of the voltage in LV, causing it to remain constant at 1 p.u, as can be observed by the orange curve.

In MV, the load voltage is the same both using LFT and SST, since neither the load nor the SST does not demand reactive from the system, thus there are no gains or losses in the voltage levels. However, the control of the SST acts during the period in which the voltage violation occurs, raising it to approximately 0.94 pu, as can be observed in Fig. 6(b).

Analyzing the power provided by the substation, it is verified that it is practically the same, both using LFT and SST, which is expected. However, when the SST control is triggered to correct the voltage, there is an increase in the active power release of the substation, due to the increase in voltage of the load (Fig. 7(a)).

Analyzing the Fig. 7(b) it is possible to notice the moments in which the SST control acts, supplying or absorbing reactive power. To correct the voltage violation, the SST injects up to around 25 kvar/phase.

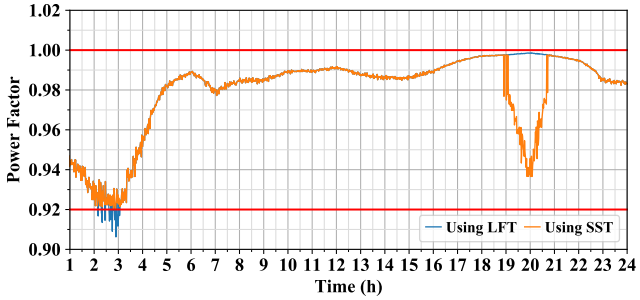


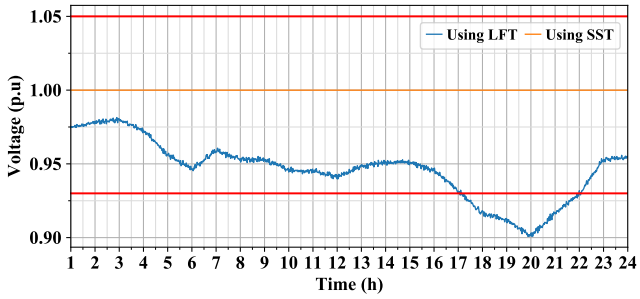
Fig. 8. Substation Power Factor

The Fig. 8 shows the power factor at the substation. Note that the SST control operates at the initial hours to correct the power factor of the substation. This action can also be observed in Fig. 7(b), where the SST makes a small adjustment in the reactive power to be able to control the power factor.

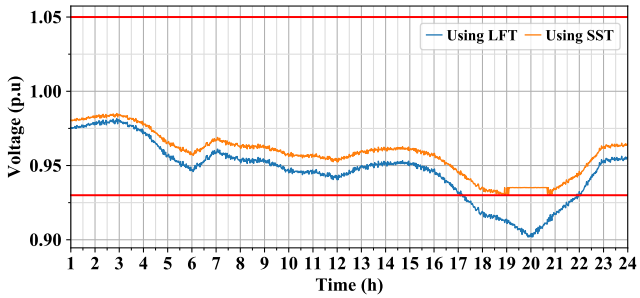
B. *Test 2*

The test 2 consists of using a load of 200 kVA with power factor of 0.9 inductive. The Fig. 9(a) shows the LV voltage, with both LFT and SST. In this case the voltage violation occurs in a longer period, from 5:30 p.m. to 10:00 p.m.

By analyzing the MV voltages, it is possible to verify an increase in the voltage level at the point of connection of the SST, simply because the SST does not demand reactive. In the period from 7:00 p.m. to 10:30 p.m. the SST starts to inject reactive power in the network, causing the voltage to be increased above the allowed limit, as observed in the orange curve of Fig. 9(b).

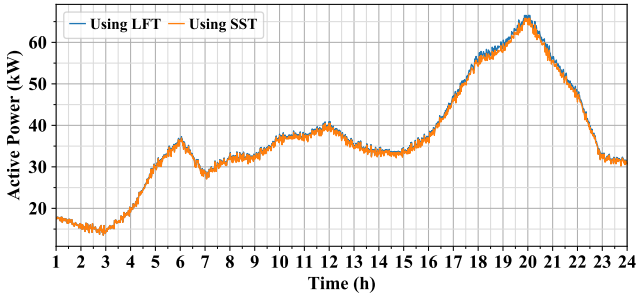


(a)

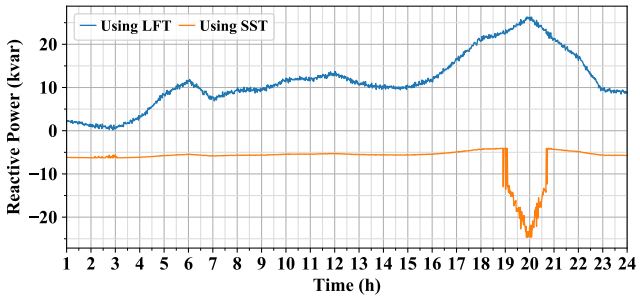


(b)

Fig. 9. Load Voltage. (a) Low Voltage. (b) Medium Voltage.



(a)



(b)

Fig. 10. Substation. (a) Active Power. (b) Reactive Power.

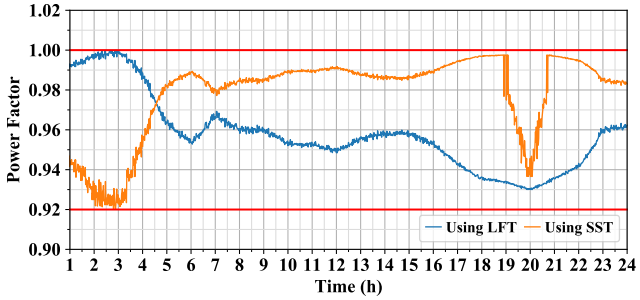


Fig. 11. Substation Power Factor

Regarding to the active power supplied by the substation, it suffers a small reduction, since the total apparent power demanded by the SST is lower than when a LFT is used, because in normal operation the SST does not demand reactive power causing a reduction in the current demanded.

In the Fig. 10(b) the difference in the use of SST and LFT is observed. Using a LFT, the reactive power demanded by the load is supplied directly by the substation. Using the SST, the reactive power of the load is supplied by the internal capacitors, so the SST does not demand reactive power of the substation. In the Fig. 10(b) it is also possible to observe the instants in which the SST control acts by injecting reactive power in order to correct the voltage levels of the load in MV.

The Fig. 11 shows the behavior of the power factor of the substation. In this case there was no violation of the power factor with LFT or with SST, however it is possible to notice the variation of this when the voltage control is triggered.

C. Test 3

The test 3 consists of using the same load of test 2, but with a penetration of 20% of PV system. The power of the PV system is defined as 20% of the rated power of the load. So, P_{PV} is equal to 40 kVA.

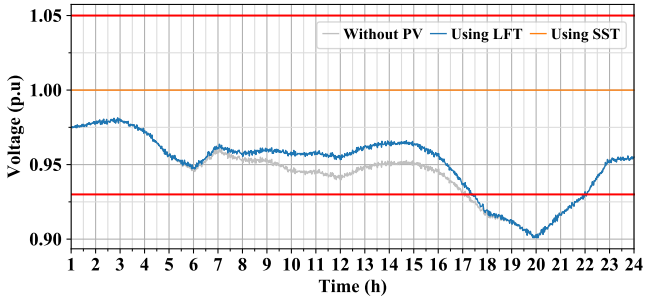
Although the PV system increases the voltage level in the period in which it is in operation, there is still a voltage violation in the night. In the Fig. 12(a) can be observed the voltage gain obtained when using the PV system only with the LFT. When using the SST there is no difference in the voltage level in LV, since it is constant in 1 p.u.

In MV, the PV system contributes to an even higher voltage increase in conjunction with the SST. In the Fig. 12(b) it can be noticed the increase of the voltage in MV, as well as the action of the control in the night period to correct the voltage level.

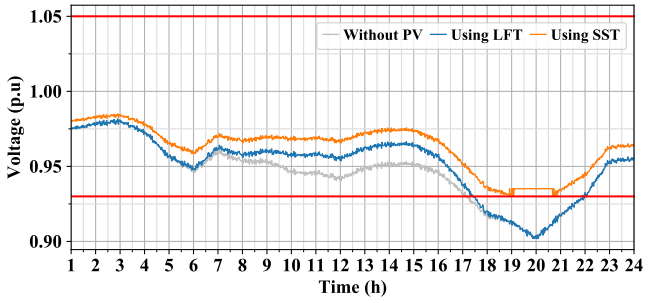
Regarding the use of PV system with LFT or SST, there are no major differences in active power provided by the substation between them. However, when a PV system is used, the power supplied by the substation presents a considerable reduction, during the period in which the PV system is in operation, since the load demand is supplied by the PV system. The difference in power supplied by the substation, with and without PV system, can be observed in Fig. 13(a).

The supply of reactive power by the substation, with or without PV, is not affected, since PV system only provides active power to the load, so Fig. 13(b) has the same characteristic as Fig. 10(b).

An important point to be considered when using PV systems is a strong decrease in the substation power factor, since the PV system causes a reduction in the active power provided by the substation, while the reactive power remains the same. In the Fig. 14 it is verified the impact that the PV system causes on the power factor when it is used with a LFT, in the period in which it is operating. Using the PV system with the SST, it

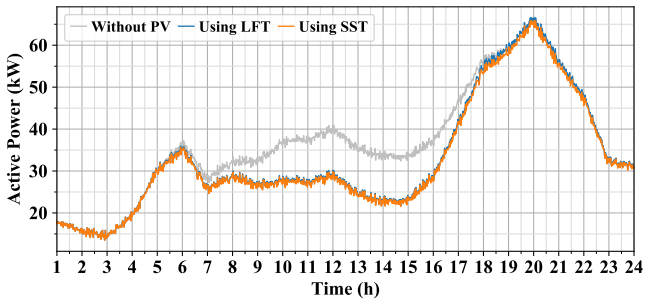


(a)

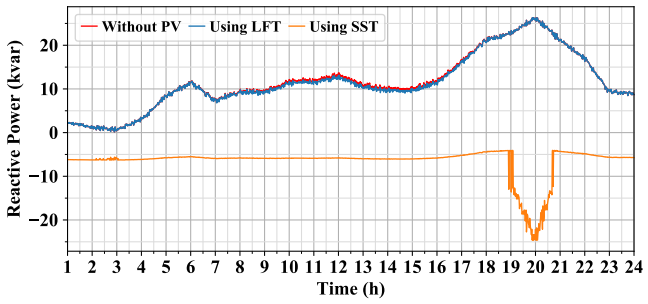


(b)

Fig. 12. Load Voltage. (a) Low Voltage. (b) Medium Voltage.



(a)



(b)

Fig. 13. Substation. (a) Active Power. (b) Reactive Power.

is noticed that the power factor presents the same behavior of test 2, however the SST control acts during the PV operation to correct the power factor violation occurred at 12:00 p.m.

V. Conslusion

With the proposed methodology, a functional model can be developed for the SST, in order to use it in the OpenDSS® platform to validate the functionalities of this equipment. In addition, using the PQ plan, it was possible to define operating limits for the SST in order to make the necessary compensation without exceeding its operational limits.

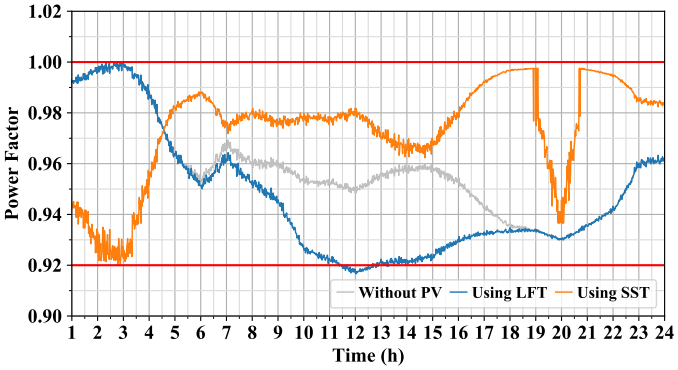


Fig. 14. Substation Power Factor

The control strategy used provided the necessary adjustments of voltage and power factor, except when the SST reaches its operating limit, not making the necessary adjustments. However, although this equipment performs several functions present in other equipment satisfactorily, it alone is not able to meet all the needs of a large network.

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